Factory Reallocation & Shipping Optimization Recommendation System

Case Study: Nassau Candy Distributor | Machine Learning Internship Project

Executive Summary

This project develops a decision-intelligence prototype for Nassau Candy Distributor to support factory-product reassignment and shipping optimization. The system combines historical order data, predictive machine learning, regional distance analysis, scenario simulation, and a Streamlit dashboard. Users can select a product, destination region, shipping mode, and optimization priority to compare factory options and receive a recommendation.

The implementation was designed as an internship deliverable and a working analytical prototype rather than a production logistics system. The source dataset contains 10,194 order records and 18 fields. Three regression models were evaluated for shipping lead-time prediction: Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor. Gradient Boosting produced the lowest RMSE and the highest R2 in the evaluated setup, although the R2 value was low, indicating that the available features explain only a small portion of observed lead-time variation.

1. Background and Business Problem

The project brief states that Nassau Candy currently assigns products to factories using static rules and legacy processes. The resulting business concerns include suboptimal shipping distances, high lead times for certain regions, and margin erosion caused by logistics inefficiencies. The requested analytical system is intended to simulate factory-product reassignment scenarios, quantify potential operational impact, and recommend better configurations at scale.

> The central decision question is: which factory assignment is a better option for a product and destination scenario when shipping efficiency and financial stability must both be considered?

2. Dataset and Data Preparation

The supplied Nassau Candy Distributor dataset contains 10,194 rows and 18 columns. Important fields include Order Date, Ship Date, Ship Mode, Customer ID, City, State/Province, Postal Code, Division, Region, Product ID, Product Name, Sales, Units, Gross Profit, and Cost. The project instructions also provide five factory coordinates and a product-to-factory mapping.

Property	Value
Records	10,194
Columns	18
Products	15
Customers	5,044
Ship modes	4
Regions	4
Factories	5
Missing values	0
Duplicate rows	0

Feature engineering included derived Lead Time Days from Ship Date minus Order Date, order-month and day-of-week features, current factory assignment, customer location labels, and profit margin. Product-to-factory mapping was taken from the project specification.

3. Predictive Modeling

The predictive objective specified in the brief is expected shipping lead time based on product, origin factory, destination region, and shipping mode. The implementation evaluates three models: Linear Regression as a baseline, Random Forest Regressor, and Gradient Boosting Regressor. Categorical features were one-hot encoded and numerical features were passed through the preprocessing pipeline. The data was split into 80% training and 20% testing samples.

Model	MAE	RMSE	R2
Gradient Boosting	214.871	260.524	0.040
Linear Regression	212.601	264.191	0.013
Random Forest	219.800	270.690	-0.036

Gradient Boosting was selected as the best-performing model in the evaluated configuration because it achieved the lowest RMSE and highest R2. The MAE of Linear Regression was slightly lower, so model selection was based on the combined evaluation objective rather than a single metric.

4. Scenario Simulation and Recommendation Logic

The system identifies the current factory for the selected product and compares it with all five available factories. Because the source dataset contains factory coordinates but does not contain customer latitude/longitude, the prototype uses representative coordinates for the four destination regions (Atlantic, Pacific, Interior, Gulf) to estimate factory-to-region distance. This creates a consistent scenario-comparison layer without inventing row-level customer coordinates.

The recommendation score combines shipping-efficiency signals and a financial-stability signal. A Speed versus Profit priority control changes the relative weight of the recommendation components. Existing product-level profit margin is used as a financial stability indicator because the dataset does not provide factory-specific freight cost or factory-specific profit data.

5. Streamlit Web Application

The final prototype is deployed as a Streamlit web application. The dashboard provides a product selector, region selector, ship-mode selector, and optimization-priority slider. It presents current factory, product sales, gross profit, profit margin, recommendation, factory comparison, what-if analysis, risk and impact messaging, ML model performance, and a dataset overview.

- Factory Optimization Simulator - compare all available factories for a selected scenario.
- What-If Scenario Analysis - compare current assignment with recommended alternatives.
- Recommendation Dashboard - rank factory options using the project scoring logic.
- Risk & Impact Panel - highlight financial-margin status and optimization warnings.
- ML Model Performance - show MAE, RMSE, and R2 for evaluated regression models.

6. Key Performance Indicators

KPI	Purpose

Lead Time Reduction (%)	Represents potential operational improvement.
Profit Impact Stability	Uses product historical margin as the available financial-stability indicator.
Scenario Confidence Score	Represents the reliability of the recommendation workflow as a decision-support prototype.
Recommendation Coverage	Measures the breadth of product/region scenarios the tool can analyze.

7. Limitations and Responsible Interpretation

- The observed lead-time target derived from the supplied dates is not strongly explained by the available predictor variables; the resulting R2 is low. The regression model should therefore be treated as a prototype signal, not a high-confidence forecasting system.
- Customer latitude/longitude is not present in the supplied dataset. Distance is therefore estimated using representative coordinates for destination regions rather than exact customer addresses.
- Factory-specific shipping/freight costs are not present. The prototype uses historical product profit margin as a financial-stability indicator instead of fabricating factory-level profit impact.
- A production deployment would require verified logistics costs, actual factory capacity, route-level shipping history, customer coordinates, service-level constraints, and business-approved optimization rules.

8. Conclusion

The project converts the Nassau Candy Distributor brief into a working decision-support prototype that combines predictive modeling, factory scenario simulation, and an interactive Streamlit dashboard. The implementation demonstrates an end-to-end machine-learning workflow: data preparation, model training and evaluation, business-oriented scenario analysis, recommendation logic, and deployment. The prototype provides a foundation that can be strengthened with richer logistics data and validated business constraints before production use.

Project Deliverables

- GitHub repository containing the notebook, Streamlit application, dependencies, and dataset.
- Live Streamlit dashboard for scenario analysis and factory recommendation.
- This research paper documenting methodology, results, insights, and limitations.
- Executive-summary information included in this report for stakeholder communication.

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